AMALA P ANTY

Data Analyst | Data Scientist | NLP & Conversational AI Specialist

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PROFESSIONAL SUMMARY

Results-driven Data Scientist and Data Analyst with 3.5+ years of experience in NLP, machine learning, and conversational AI for large-scale BFSI clients. Proven expertise in building and validating ML models, designing knowledge graphs (ontology) ,performing statistical analysis, Automation using Jenkins and delivering data-driven insights to optimize operational efficiency. Proficient in Python, SQL, Scikit-learn, and data visualization tools. Strong domain knowledge in banking and financial services with a track record of measurable performance improvements. Seeking to leverage a strong ML, NLP, and data analytics foundation in Data Analyst or Data Scientist roles.

CORE TECHNICAL SKILLS

Programming & DB	Python, SQL, Regex
ML & AI	Supervised Learning, Unsupervised Learning, Classification, Regression, PCA, XGBoost, Feature Engineering, Hyperparameter Tuning,Model Evaluation Metrics such as Confusion Matrix, Classification Report(Accuracy, Precision, Recall, F1), Streamlit
Deep Learning & NLP	ANN, CNN, RNN, Intent Classification, NER, Dialog Management, FAQ Modeling, LLM Response Validation, Chatbot Training, Sentiment Analysis
Frameworks & Libs	Scikit-learn, TensorFlow, Keras, Pandas, NumPy, Matplotlib, Seaborn
Data Analytics	EDA, Statistical Analysis, Advanced Excel, Pivot Tables, Data Wrangling, Data Cleaning
Tools & Platforms	Git, Jenkins, Postman, Jira, Sublime Text, Google Translate, Triniti
Environments	Jupyter Notebook, Visual Studio Code, Google Colab
Domain	Banking & Financial Services (BFSI), Finacle, Saksham 360,Artificial Intelligence(Chatbots)

MACHINE LEARNING PROJECTS

E-Commerce Customer Churn Prediction | *Supervised Learning | Classification*

- Built an XGBoost classifier to predict customer churn with 92.09% accuracy using Python and scikit-learn.
- Performed feature engineering, missing value treatment, and GridSearchCV-based hyperparameter tuning.
- Evaluated model with Accuracy, Precision, Recall, and F1-Score to optimize for business-critical false-negative reduction.
- Deployed as an interactive **Streamlit web app** with real-time prediction

Brain Tumor Detection using CNN | *Deep Learning | Image Classification*

- Developed a Convolutional Neural Network (CNN) to classify MRI brain scans as tumor-positive or tumor-negative using TensorFlow and Keras.
- Applied image preprocessing, data augmentation, and transfer learning by using cnn architecture MobileNetV2 to improve model generalization and robustness.
- Evaluated model performance using Accuracy, Precision, Recall, and F1-Score; achieved high classification accuracy on the test dataset.
- Built an end-to-end inference pipeline deployable via a Streamlit web application for real-time predictions.

Titanic Survival Prediction | *Supervised Learning | Classification*

- Developed a binary classification model using XGBoost with feature engineering and hyperparameter tuning, achieving 80.09% accuracy on test data and deployed as an interactive **Streamlit web app** with real-time prediction

Credit Card Customer Segmentation | *Unsupervised Learning | Clustering*

- Segmented customers using KMeans Clustering and PCA for dimensionality reduction; visualized cluster profiles with Matplotlib and Seaborn to derive actionable insights.

PROFESSIONAL EXPERIENCE

Data Engineer II | Gupshup Technologies India Pvt. Ltd.|*Jul 2021 – Dec 2024*

Clients: ICICI Bank | Canara Bank | Standard Chartered Bank | Tata Capital | Tonik

Chatbot Training & NLP Model Development

- Analyzed banking FAQs (account balance, EMI, loan tenure, card dues, transaction history) and built high-accuracy NLP training datasets for end-to-end chatbot deployment.
- Identified and corrected intent mismatches in bot responses; iteratively retrained models to achieve 100% accuracy on targeted use cases.
- Validated and scored LLM/ChatGPT-generated responses (correct/incorrect/invalid) to enhance model precision for production environments.
- Developed utterances for 20+ banking use cases (credit card cancellation, loan balance, etc.) and authored NER, Dialog, and Smalltalk configuration files.

Knowledge Graph & Ontology Management

- Designed and maintained base Knowledge Graphs for Retail Banking (India/US) and Corporate Banking generic projects from scratch.
- Structured product-attribute data across account, loan, card, deposit, and investment domains; performed FAQ fulfillment analysis and ontology updates aligned to new intents and FAQs.
- Conducted admin validation, sanity testing, and QA handover for all completed knowledge graph projects.

Chat Log Analysis & Accuracy Improvement

- Executed Python scripts to generate chat log validation reports; classified utterances by intent (FAQ, small talk, transaction, query, gibberish).
- Identified gaps, recommended new FAQs, prepared structured V4-format training data, and incorporated customer feedback for accuracy improvement loops.

Multilingual Data Validation & Curation

- Curated, updated, and deduplicated FAQ datasets for ICICI Bank, Tata Capital, and Standard Chartered Bank.
- Validated multilingual translations using Google Translate and Triniti; performed manual word-level reviews and compiled data for translation engine fine-tuning.
- Tracked issues via Jira and maintained validation trackers across multiple concurrent client projects.

Automation & CI/CD Pipeline

- Automated chatbot training and regression testing via Jenkins; generated pivot table reports for each automation cycle.
- Resolved automation defects across FAQ, PC, NER, Dialog, and Smalltalk components; designed POC data (text/image generation) and committed finalized training data to Git.

Deputy Manager – Banking Operations | Axis Bank| *Oct 2020 – Nov 2021*

- Managed end-to-end banking operations: deposits, payments, NEFT/RTGS transfers, cheque clearing, and end-of-day reconciliation.
- Processed service requests including loan closures, re-KYC updates, debit card issuance, FD receipts, and locker operations.
- Investigated and resolved ATM/debit/credit card fraud escalations; monitored transactions for suspicious activity.
- Cross-sold fixed deposits, insurance, mutual funds, and loans; drove combined FD and insurance sales exceeding Rs. 3 crores.
- Conducted onboarding training for new joiners on Finacle, Saksham 360, and core banking payment procedures.

Deputy Manager – Home Loans | ICICI Bank| *Oct 2019 – Nov 2020*

- Assessed applicants' financial status through bank statements, credit reports, and document verification to determine home loan eligibility.
- Guided customers through loan selection, application, and approval; consistently achieved monthly disbursement targets of Rs. 1.5 crores.

- Managed loan portfolios including part payments, foreclosure, rate conversions, EMI, and loan closure across home, personal, and auto loan products.
- Cross-sold credit cards, mutual funds, personal loans, and fixed deposits; secured FD bookings of Rs. 2 crores (FY 2018-2019), earning Top Performer recognition.

EDUCATION

MBA – Finance & Marketing *2016 – 2018*

Rajagiri College of Social Sciences (Autonomous), RCBS | 70%

BBA – Business Administration *2013 – 2016*

Rajagiri College of Management and Applied Sciences, RCMAS | 84%

KEY ACHIEVEMENTS

- Received Client Appreciation for educating stakeholders on Knowledge Graph architecture and chatbot testing at Gupshup Technologies.
- Consistently achieved monthly home loan disbursement target of Rs. 1.5 crores at ICICI Bank.
- Awarded Top Performer at ICICI Bank for securing Rs. 2 crores in fixed deposits during FY 2018-2019.
- Drove insurance and fixed deposit sales exceeding Rs. 3 crores at Axis Bank.
- Earned repeated client commendations for proactive follow-ups and timely execution of critical banking activities.

I hereby declare that all information furnished above is true and accurate to the best of my knowledge.